Introduction to Machine Learning

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Lecture 1: Introduction and ML Problem Framing

1.1 Learning from data

Machine learning builds systems that improve through data or experience. It is useful when desired behavior is difficult to specify as rules but examples or an evaluation of the result are available, as in spam filtering. Training uses data to fit a model; inference applies the fitted model to a new input. Artificial intelligence is a broader goal that also includes approaches such as symbolic reasoning and planning.

A model $h_w \colon X \to Y$ has parameters w learned from data or interaction. Input representation, target, model family, and evaluation criterion all affect what can be learned.

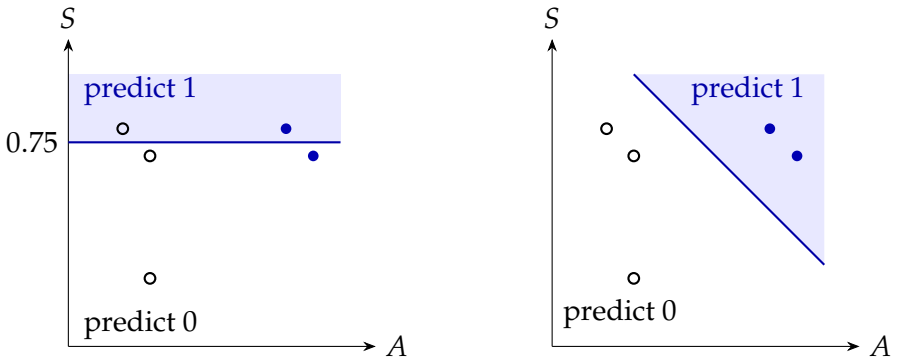
1.2 Supervised learning and decision regions

In supervised learning, training examples have inputs and targets, $\{(x_i, y_i)\}_{i=1}^{n}$. Each row is a sample; the components of x_i are features. Classification predicts a discrete class, while regression predicts a continuous target. A binary classifier partitions feature space into regions assigned to classes 0 and 1.

Consider the taste-test data below, where A is acidity, S is sweetness, and $y = 1$ means great taste. A numerical taste score would instead make this a regression task.

Beverage	1	2	3	4	5
A	0.8	0.3	0.2	0.3	0.9
S	0.8	0.25	0.8	0.7	0.7
y	1	0	0	0	1

A classifier assigns a label to every point (A, S), not just the five observed points. Its decision boundary separates regions receiving different predictions. The plots below show the same data under two rules.



(a) Stump: $S \geq 0.75$ (b) Linear: $A + S \geq 1.3$

Figure 1: Feature space for the taste test. Filled blue points have label 1; open points have label 0. Shading marks the predicted class 1 region, including its boundary. The plotted square is only a window into feature space.

A DECISION STUMP makes one threshold test. The rule $f(A, S) = \mathbf{1}\{S \geq 0.75\}$ predicts $(1, 0, 1, 0, 0)$, so its training accuracy is $3/5$. The horizontal boundary shows why: the stump ignores acidity, so the high-sweetness negative example is misclassified.

A decision tree applies tests in sequence. Each path from the root to a leaf specifies a region, and the leaf supplies its prediction. For fixed splits, assigning the majority training label to each leaf maximizes training accuracy. The tree below gets all five labels right, although its second split does not change the prediction: both of its leaves predict 1.

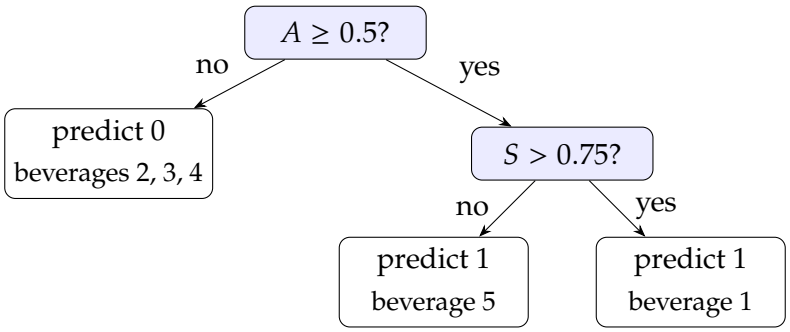


Figure 2: A depth-two tree for the same data. Follow the answer at each test to reach a prediction. Here the first split alone already separates the labels.

A LINEAR CLASSIFIER uses a straight decision boundary. For $f(A, S) = \mathbf{1}\{A + S - 1 \geq 0\}$, the boundary is $A + S = 1$ and the predictions are $(1, 0, 1, 1, 1)$, giving $3/5$ accuracy. Raising the threshold to 1.3 shifts the parallel boundary and gives $(1, 0, 0, 0, 1)$, fitting all five. The rule predicts 1 on the boundary as well as above it. Unlike a tree's axis-aligned tests, this rule trades off acidity and sweetness along a slanted boundary.

1.3 Generalization and evaluation

The aim is to predict for new inputs, not merely reproduce training labels. For a classifier, training accuracy is the fraction of training predictions that match their labels; 0–1 loss is the fraction that do not. A held-out test set assesses performance on examples not used to fit or select the model. The evaluation is informative about future use only to the extent that those examples represent the intended deployment setting.

A lookup rule can memorize every training example and output zero elsewhere. If identical feature vectors have consistent labels, it has perfect training accuracy, yet its predictions on unseen points need not be useful. In particular, it fails on unseen positive examples; its actual test accuracy depends on the test distribution. Good training fit alone therefore does not establish GENERALIZATION. Simpler models can express useful assumptions about unseen data, but simplicity alone does not guarantee success.

1.4 Other learning settings

UNSUPERVISED LEARNING uses inputs without explicit target labels. Clustering groups similar observations; density estimation models how data are distributed; dimensionality reduction seeks a lower-dimensional representation that retains useful structure. Other examples include anomaly detection and data generation. A reduced representation can support visualization or subsequent learning tasks.

REINFORCEMENT LEARNING learns behavior from rewards, often over a sequence of actions. Unlike a labeled input-output example, the reward may evaluate a sequence or outcome, so choosing actions requires relating feedback to earlier decisions.

1.5 The machine-learning lifecycle

1. **Frame the learning problem.** Specify the target and objective, choose a measure of success, identify available data and features, and split training and test data.
2. **Design the model.** Choose a model family or architecture and its hypothesis space. Its inductive biases are assumptions that favor some solutions over others.
3. **Optimize.** Define a loss and choose an optimization method to fit the model. Manage regularization and overfitting.
4. **Predict and evaluate.** Apply the model and measure performance with metrics suited to the objective. Check whether the model works for the intended reasons and population, and use the results to revisit earlier choices.

Source. Jennifer Listgarten and Alex Dimakis, Introduction to Machine Learning, Spring 2026, Lecture 1.